where we have written U'' for the Hessian matrix of U and g_k'' for the Hessian matrix of g_k.

For simplicity, we will assume the mass matrix is diagonal with ith diagonal element m_i. Now we write a formula for the time rate of change of the wedge product

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{i=1}^{\mathrm{N}_c}\mathrm{d}q_i\wedge\mathrm{d}p_i=\sum_{i=1}^{\mathrm{N}_c}\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}q_i\wedge\mathrm{d}p_i+\mathrm{d}q_i\wedge\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}p_i,$$

then use formulas (4.11)–(4.13) to simplify, obtaining

$$\sum_{i=1}^{\mathrm{N}_c}\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}q_i\wedge\mathrm{d}p_i=\sum_{i=1}^{\mathrm{N}_c}m_i^{-1}\mathrm{d}p_i\wedge\mathrm{d}p_i=0,$$

and, writing $U_{ij}''=\partial^2U/\partial q_i\partial q_j$,

$$\begin{aligned}\sum_{i=1}^{\mathrm{N}_c}\mathrm{d}q_i\wedge\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}p_i=&-\sum_{i=1}^{\mathrm{N}_c}\sum_{j=1}^{\mathrm{N}_c}\mathrm{d}q_i\wedge U_{ij}''\mathrm{d}q_j\\&-\sum_{i=1}^{\mathrm{N}_c}\sum_{k=1}^{l}[\mathrm{d}q_i\wedge(\mathrm{d}\lambda_k)g_{ki}'+\mathrm{d}q_i\wedge\lambda_k\sum_{j=1}^{\mathrm{N}_c}(g_k'')_{ij}\mathrm{d}q_j].\end{aligned}$$

Treating each term in turn, first observe that

$$\sum_{i=1}^{\mathrm{N}_c}\sum_{j=1}^{\mathrm{N}_c}\mathrm{d}q_i\wedge U_{ij}''\mathrm{d}q_j=0.$$

This follows from the antisymmetry of the wedge product ($\mathrm{d}x\wedge\mathrm{d}y=-\mathrm{d}y\wedge\mathrm{d}x$), and the fact that the Hessian matrix of U is symmetric. Next, note that

$$\sum_{i=1}^{\mathrm{N}_c}\sum_{k=1}^{l}\mathrm{d}q_i\wedge\lambda_k\sum_{j=1}^{\mathrm{N}_c}(g_k'')_{ij}\mathrm{d}q_j=\sum_{k=1}^{l}\lambda_k\sum_{i=1}^{\mathrm{N}_c}\mathrm{d}q_i\wedge\sum_{j=1}^{\mathrm{N}_c}(g_k'')_{ij}\mathrm{d}q_j=0,$$

by the same reasoning, since each of g_k'' is a symmetric matrix. Finally

$$\sum_{i=1}^{\mathrm{N}_c}\sum_{k=1}^{l}[\mathrm{d}q_i\wedge(\mathrm{d}\lambda_k)g_{ki}'=\sum_{k=1}^{l}\left(\sum_{i=1}^{\mathrm{N}_c}g_{ki}'\mathrm{d}q_i\right)\wedge\mathrm{d}\lambda_k=\sum_{k=1}^{l}\mathrm{d}g_k\wedge\mathrm{d}\lambda_k=0,$$

since we know that $\mathrm{d}g_k=0$ from (4.13). Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{i=1}^{\mathrm{N}_c}\mathrm{d}q_i\wedge\mathrm{d}p_i=0.$$